Decision and Risk Analysis

Lecture 11: **Evaluating Regression Models**

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Alexander Abuabara



Roadmap

Q1: Is the model fair?

Q2: How wrong is the model?

Q3: How accurate are the predictions?

Normal Hierarchical Models

Practice Questions

Why evaluate a model?

Analogy

You would not eat a foraged mushroom or ride in a home-built car without checking it is safe. The same holds for models.

Three critical questions, for *any* model (Bayesian or frequentist, simple or big):

1. **How fair** is the model?
2. **How wrong** is the model?
3. **How accurate** are the posterior predictive models?

Running example: Capital Bikeshare (Washington, DC)

$$Y_i \mid \beta_0, \beta_1, \sigma \overset{ind}{\sim} N(\mu_i, \sigma^2), \quad \mu_i = \beta_0 + \beta_1 X_i$$
$$\beta_{0c} \sim N(5000, 1000^2), \;\; \beta_1 \sim N(100, 40^2), \;\; \sigma \sim \text{Exp}(0.0008)$$

Y = daily rides, X = temperature (feels-like, °F), $n = 500$ days.

Question 1: Is the model fair?

Ask, *every time*, even when the answer seems uneventful:

- How was the data collected?
- By whom, and for what purpose?
- How might the analysis or data collection impact individuals and society?
- What biases or power structures might be baked in?

Bikeshare check

Data come only from people who use the service. The model may better serve *this* population at the exclusion of others. No perspective is neutral.

Models are not always fair

- **Hiring:** a résumé-screening model trained on mostly male employees reinforced a preference for male applicants.
- **Facial recognition:** trained on unrepresentative images, so misidentification is more common for under-represented groups.
- **Risk classification:** a detention/release model that recommended detention in nearly all cases.

Discussion

Which of the four fairness questions would have caught each of these problems earlier?

Question 2: How wrong is the model?

"All models are wrong, but some are useful." (George Box)

The right question is not *is* it wrong, but *how* wrong. Check the three assumptions of $Y_i \mid \beta_0, \beta_1, \sigma \overset{ind}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2)$:

1. **Independence:** given X, cases are independent.
2. **Linearity:** typical Y is a linear function of X.
3. **Normality and constant variance:** Y varies normally around μ with the same σ at every X.

Bikeshare

Ridership is correlated over time, but temperature explains much of that. Given X, independence is "good enough to proceed."

Posterior predictive check: the idea

If assumptions are reasonable, the posterior model should be able to **simulate data that looks like the observed data**.

For each of the $N = 20{,}000$ MCMC draws $\theta^{(j)} = (\beta_0^{(j)}, \beta_1^{(j)}, \sigma^{(j)})$ simulate $n = 500$ outcomes:

$$Y_i^{(j)} \sim N\left(\beta_0^{(j)} + \beta_1^{(j)} X_i,\ (\sigma^{(j)})^2\right)$$

$$\begin{bmatrix} \theta^{(1)} \\ \vdots \\ \theta^{(N)} \end{bmatrix} \longrightarrow \begin{bmatrix} Y_1^{(1)} & \cdots & Y_{500}^{(1)} \\ \vdots & & \vdots \\ Y_1^{(N)} & \cdots & Y_{500}^{(N)} \end{bmatrix}$$

Rows = simulated datasets (used here). **Columns** = posterior predictive model for one day (used later).

Posterior predictive check in R

```
library(bayesrules); library(rstanarm); library(bayesplot)

# bike_model <- stan_glm(rides ~ temp_feel, data = bikes,
#   family = gaussian,
#   prior_intercept = normal(5000, 1000),
#   prior = normal(100, 40),
#   prior_aux = exponential(0.0008),
#   chains = 4, iter = 5000*2, seed = 84735)

# Compare 50 of the 20,000 simulated datasets to the data
pp_check(bike_model, nreps = 50) + xlab("rides")
```

- **Dark curve** y: the observed data. **Light curves** y_{rep}: simulated datasets.
- **Bikes result:** center and range are captured, but most simulations miss the **bimodality**. Not a bad model, just one that could be *less wrong*.

Reading a `pp_check()`

Good fit

- Light curves wrap tightly around the dark curve
- Similar center, spread, and shape

Poor fit

- Dark curve skewed, light curves symmetric
- Simulated Y has wrong center or too much/little variability
- Missing modes, heavy tails, impossible values (e.g., negative counts)

Quick intuition

If you asked the model to "write" 50 fake datasets and a detective could instantly pick out the real one, the model is missing something.

Dealing with wrong models

Assumption 1 fails (dependent data)

- Repeated measures on the same subject (e.g., 3 memory tests per adult) $\Rightarrow$ **hierarchical models** (Unit 4, revisited at the end).
- Time/space dependence $\Rightarrow$ time series, spatial, spatiotemporal models.

Assumptions 2 and 3 fail (nonlinear, non-constant variance)

- **Different data structure:** Poisson, Negative Binomial, Binomial (Ch. 12–13).
- **Transform:** $g(Y)$, $h(X)$, or both, e.g. $\log(Y)$, $\sqrt{X}$, X^2.

$$\log(Y_i) \mid \beta_0, \beta_1, \sigma \overset{ind}{\sim} N(\beta_0 + \beta_1 X_i, \sigma^2)$$

Example: Y is right-skewed and its spread grows with X. After $\log(Y)$, the relationship is linear, the spread is constant, and the `pp_check()` looks right.

Question 3: How accurate are the posterior predictions?

A good model **generalizes** beyond the data used to build it. Three tools, from simplest to most technical:

1. Posterior predictive summaries (MAE, scaled MAE, within-50/95)
2. Cross-validation (honest, out-of-sample versions of the same summaries)
3. Expected log-predictive density (ELPD) for comparing models

One case: October 22, 2012

Temperature $= 75°$F, observed $Y = 6228$ rides.

```
predict_75 <- bike_model_df %>%
  mutate(mu = `(Intercept)` + temp_feel*75,
         y_new = rnorm(20000, mean = mu, sd = sigma))
```

Posterior predictive mean Y'	3967
Prediction error $Y - Y'$	2261
Posterior sd	1280
Scaled error $(Y - Y')/\text{sd}$	1.767
50% interval	(3096, 4831) $\rightarrow$ **outside**
95% interval	(1500, 6482) $\rightarrow$ **inside**

The model under-predicted, but the outcome was still within the realm of possibility.

Why scale the error?

Two models both predict $Y' = 3967$ and observe $Y = 6228$ (error $= 2261$ for both).

	Our model	Alternative
Posterior sd	1280	500
Absolute error	2261	2261
Scaled error	**1.77**	**4.5**

Takeaway

The same absolute error is far more damning when the model claimed to be confident. Scaling by the posterior sd accounts for the model's *own* stated uncertainty. As a rule of thumb, $|\text{scaled error}| > 2$ or 3 signals a poor prediction (for roughly symmetric posteriors).

Posterior predictive summaries (all n cases)

Each Y_i has a posterior predictive mean Y_i' and sd_i.

- **MAE** (median absolute error): $\text{median}\,|Y_i - Y_i'|$
- **Scaled MAE:** $\text{median}\,\dfrac{|Y_i - Y_i'|}{\text{sd}_i}$
- **within_50 / within_95:** proportion of observed Y_i inside their 50% / 95% posterior prediction intervals.

Calibration check: within_50 should be near 0.50 and within_95 near 0.95.

`stable = TRUE` switches to the median and median absolute deviation, which is useful for skewed posteriors.

Bikeshare: summaries on the training data

```
set.seed(84735)
predictions <- posterior_predict(bike_model, newdata = bikes
    )
dim(predictions)    # 20000 x 500
ppc_intervals(bikes$rides, yrep = predictions,
              x = bikes$temp_feel, prob = 0.5, prob_outer =
                  0.95)

prediction_summary(bike_model, data = bikes)
```

mae	mae_scaled	within_50	within_95
989.7	0.7712	0.438	0.968

Typically, ridership is about 990 rides (0.77 sd) from its prediction. 43.8% of days fall in their 50% interval and 96.8% in their 95% interval. Is 990 good enough? **It depends on context** (ask Capital Bikeshare).

The problem: testing on the data you trained on

The anchov-lada analogy

You build every taco recipe around Reem, who loves anchovies. Her approval does not mean the general public will like it.

A model is optimized to the data it was built on, so evaluating accuracy on that *same* data is **overly optimistic**.

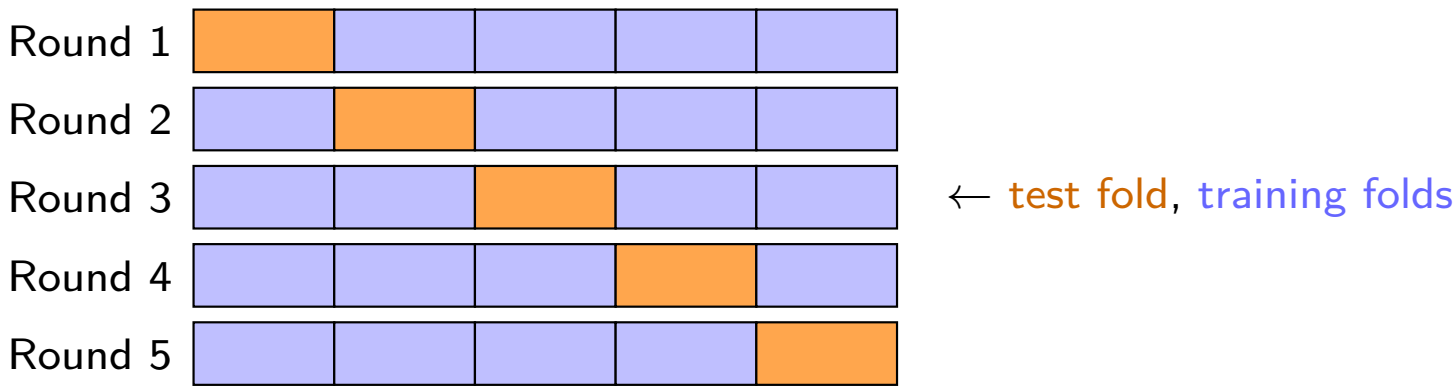
Idea: split the existing data into roles.

- **Train:** build the model on, say, 90% (450 days).
- **Test:** predict the held-out 10% (50 days) and compute MAE, etc.

One split is **unstable**: a different random split may paint a different picture.

k-fold cross-validation

1. **Create folds:** split data into k non-overlapping folds of roughly equal size.
2. **Train and test:** train on $k-1$ folds, test on the remaining one, record MAE etc.
3. **Repeat** $k-1$ more times, each time holding out a different fold.
4. **Average** the k measures into one cross-validation estimate.



Choosing k (Goldilocks): $k = 2$ trains on too little data (pessimistic). $k = n$ is leave-one-out (LOO): needs n models (slow). $k = 10$ is a common compromise.

Cross-validation in R and results

```
set.seed(84735)
cv_procedure <- prediction_summary_cv(
  model = bike_model, data = bikes, k = 10)
cv_procedure$folds   # metrics per fold
cv_procedure$cv      # averaged estimate
```

	mae	mae_scaled	within_50	within_95
In-sample	989.7	0.7712	0.438	0.968
10-fold CV	1029	0.8015	0.420	0.968

Per-fold MAE ranged from about 787 to 1271. To predict tomorrow's ridership, expect to be off by about **1029 rides**, not 990. In-sample summaries gave a rosier picture.

Expected log-predictive density (ELPD)

The posterior predictive pdf for a new day:

$$f(y' \mid \vec{y}) = \int f(y' \mid \theta)\, f(\theta \mid \vec{y})\, d\theta$$

The higher $f(y_{\text{new}} \mid \vec{y})$ (equivalently its log) at the value that actually occurs, the better the prediction.

ELPD

The average of $\log f(y_{\text{new}} \mid \vec{y})$ over possible new data points.
Higher is better.

Computing and using ELPD

```
model_elpd <- loo(bike_model)    # leave-one-out CV
model_elpd$estimates
#           Estimate       SE
# elpd_loo  -4289.0 13.1186
# p_loo         2.5  0.1638
# looic      8578.1 26.2371
```

- ELPD $= -4289$ is hard to interpret alone; its scale changes from analysis to analysis.
- Its power is in **comparing models fit to the same data**:

```
loo_compare(loo(model_1), loo(model_2))  # larger elpd wins
```

Improving posterior predictive accuracy

1. **Collect more data.** It cannot strengthen a weak Y–X relationship, but it improves our understanding of it.
2. **Use different or more predictors.** Maybe humidity beats temperature, or use both (Chapter 11). There are limits to this strategy.

Two different questions

How good is the MCMC simulation? (trace plots, $\hat{R}$, effective sample size)
How good is the model? (fair, not too wrong, accurate)
You can approximate a good model badly, or approximate a bad model well.

Summary: the evaluation checklist

Question	Tool	Look for
Fair?	Context, ethics review	No harmful bias, appropriate use
Wrong?	`pp_check()`	Simulated data resemble observed data
Accurate?	`prediction_summary()`, `prediction_summary_cv()`, `loo()`	Small MAE, calibrated within_50/95, higher ELPD

Always prefer **cross-validated** metrics when judging how well a model will predict new data.

Why a hierarchical model?

Grouped data: observations within groups $j = 1, \ldots, J$ are correlated (violating independence), e.g. several songs per artist, several tests per student.

Complete pooling
Ignore groups; one mean μ for everyone.
Ignores group differences.

No pooling
Separate mean μ_j per group, estimated in isolation.
Noisy for small groups; cannot generalize.

Partial pooling
Group means share a common distribution.
Borrows strength across groups.

The Normal hierarchical model (no predictors)

$$
\begin{aligned}
Y_{ij} \mid \mu_j, \sigma_y &\sim N(\mu_j, \sigma_y^2) && \text{within-group (layer 1)} \\
\mu_j \mid \mu, \sigma_\mu &\sim N(\mu, \sigma_\mu^2) && \text{between-group (layer 2)} \\
\mu,\ \sigma_y,\ \sigma_\mu &\sim \text{priors} && \text{global parameters (layer 3)}
\end{aligned}
$$

- μ: global average. μ_j: group-specific average.
- σ_y: **within-group** variability. σ_μ: **between-group** variability.

Intraclass correlation

The fraction of total variance due to group differences:

$$\text{ICC} = \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_y^2}$$

Evaluating a hierarchical model: the same three questions

1. **Fair?** Are all groups represented? Are small or under-represented groups hurt by shrinkage toward the overall mean?
2. **Wrong?** Use `pp_check()`: simulate Y_{ij} through *both* layers and compare with the data. Also check whether the Normal assumption for the μ_j is plausible (look at group-level estimates).
3. **Accurate?** Posterior predictive summaries, **cross-validation** (done carefully), and ELPD for comparing pooling strategies.

Step 1: Is it wrong? Posterior predictive check

```
# Hierarchical model with only a group-level intercept
hier_model <- stan_glmer(
  y ~ (1 | group), data = dat, family = gaussian,
  chains = 4, iter = 5000*2, seed = 84735)

pp_check(hier_model, nreps = 50)
```

- Does the overall shape of y_{rep} match y?
- Also check by group: if a group's data consistently sits outside its simulations, the common $N(\mu, \sigma_\mu^2)$ for group means (or constant σ_y) may be wrong.
- Compare σ_y vs. σ_μ to see where the variability lives (ICC).

Step 2: Accuracy: two kinds of predictions

(a) A new case from an existing group j

$$Y_{\text{new},j} \mid \vec{y} \sim N(\mu_j, \sigma_y^2)$$

using the posterior for *that group's* μ_j (narrower, group-specific).

(b) A case from a brand-new group

$$\mu_{\text{new}} \sim N(\mu, \sigma_\mu^2), \quad Y \sim N(\mu_{\text{new}}, \sigma_y^2)$$

Uncertainty includes σ_μ (wider).

Evaluate each scenario separately. Predicting a new group is harder and is the more honest test of generalization.

Cross-validation for grouped data

Folds must respect the question you care about.

```
# In-sample summaries (existing groups)
set.seed(84735)
prediction_summary(hier_model, data = dat)

# New groups: hold out whole groups at a time
prediction_summary_cv(model = hier_model, data = dat,
                      group = "group", k = 10)

# ELPD (LOO) to compare pooling strategies
loo_compare(loo(complete_pool), loo(no_pool), loo(hier_model
    ))
```

Compare complete, no, and partial pooling by cross-validated MAE and ELPD. Partial pooling typically wins when groups differ but have limited data each.

Shrinkage and the bias–variance trade-off

The hierarchical estimate of a group mean is a weighted average:

$$\hat{\mu}_j \approx \underbrace{w_j\,\bar{y}_j}_{\text{group data}} + \underbrace{(1-w_j)\,\hat{\mu}}_{\text{global mean}}, \qquad w_j = \frac{n_j/\sigma_y^2}{n_j/\sigma_y^2 + 1/\sigma_\mu^2}$$

- Small n_j or large σ_y: w_j small, **more shrinkage** toward $\hat{\mu}$.
- Large n_j or large σ_μ: w_j large, **less shrinkage**.

Shrinkage adds a little **bias** but reduces **variance**, which usually improves out-of-sample prediction.

Worked example: three classrooms

Test scores; global mean $\hat{\mu} = 75$, $\sigma_y = 10$, $\sigma_\mu = 5$.

Classroom	n_j	$\bar{y}_j$	w_j	$\hat{\mu}_j$
A	4	90	0.50	82.5
B	25	90	0.86	87.9
C	100	90	0.96	89.4

Same raw average (90), different shrinkage. Evidence for A is thin, so its estimate is pulled strongly toward 75; C barely moves.

Why it helps prediction

A small class that scored 90 by luck will likely score lower next time. The shrunken estimate anticipates this and gives better future predictions.

Hierarchical evaluation checklist

- **Fair:** no group systematically disadvantaged by pooling or by who is in the data.
- **Wrong:** `pp_check()` overall and by group; sanity-check normality of group effects and constant σ_y.
- **Accurate:**
  - Separate summaries for *existing-group* and *new-group* prediction.
  - Cross-validation that holds out observations or whole groups.
  - ELPD (`loo`) to compare complete, no, and partial pooling.
- **Remember:** separately check MCMC quality (R-hat, effective sample size). Hierarchical posteriors can be harder to sample.

Practice Questions (1/2)

1. Name the three big questions for evaluating any model.
2. In a `pp_check()`, what does the dark curve represent, and what does each light curve represent?
3. Model A and Model B both predict $Y' = 50$ and observe $Y = 60$. Model A has posterior sd $= 2$; Model B has sd $= 10$. Which model predicted better, and why?
4. Your model's cross-validated MAE is larger than its in-sample MAE. Is that surprising? Why?

Practice Questions (2/2)

5. A model has within_95 $= 0.70$. What does that suggest?
6. Why is a single ELPD value hard to interpret, and what is it best used for?
7. In 10-fold CV with $n = 500$, how many models are trained and how many points are in each training set?
8. In a Normal hierarchical model, a group has $n_j = 3$ observations. Will its estimate be shrunk more or less than a group with $n_j = 300$?
9. When evaluating a hierarchical model, why evaluate "new group" prediction separately from "existing group" prediction?

Quick Answers (1/2)

1. **Fair? Wrong? Accurate?** (data/context/ethics; assumptions via posterior predictive check; predictive accuracy of the posterior predictive models).
2. Dark curve: the observed data y. Each light curve: one dataset y_{rep} simulated from the posterior model at one parameter draw.
3. **Model B.** Absolute errors are equal (10), but scaled errors are $10/2 = 5$ for A and $10/10 = 1$ for B. Model A is overconfident: the observation is 5 sds from its prediction, while B's predictive model comfortably covers it.
4. Not surprising. In-sample evaluation is optimistic because the model was fit to those same points; cross-validation gives a more honest estimate of error on new data.

Quick Answers (2/2)

5. The 95% prediction intervals are too narrow: only 70% of observations fall inside, instead of about 95%. The model is overconfident, so assumptions (e.g. variance or shape) may be wrong.
6. The ELPD scale depends on the data and model, so no universal "good" value exists. Use it to **compare models** fit to the same data; higher is better.
7. 10 models; each trained on $9/10 \cdot 500 = 450$ points and tested on 50.
8. **More** shrinkage toward the global mean, since little data about that group means the weight w_j on its own sample mean is small.
9. Predicting a new group also requires uncertainty about the group mean (σ_μ), so it is harder and has wider intervals. Accuracy for existing groups can look good while generalization to new groups is poor.